COBOL: Currency Denomination Calculator

1. AIM

To develop a COBOL program that accepts a total amount, validates that the amount is a multiple of 100, and calculates the number of ■500, ■200, and ■100 notes required to represent the amount. If the amount is not a multiple of 100, the program displays an invalid amount message.

2. PROBLEM STATEMENT

Write a COBOL program to accept a total amount from the user. Check whether the entered amount is divisible by 100. If it is not divisible by 100, display “NOT VALID AMOUNT” and terminate the program. Otherwise, calculate the maximum possible number of ■500 notes, then ■200 notes, and finally ■100 notes. Display the number of notes for each denomination.

3. ALGORITHM

Step 1: Start the program.

Step 2: Declare variables to store the total amount and the counts of ■500, ■200, and ■100 notes.

Step 3: Accept the total amount from the user.

Step 4: Check whether FUNCTION MOD(TOTALAMOUNT, 100) is not equal to zero.

Step 5: If the condition is true, display “NOT VALID AMOUNT” and stop the program.

Step 6: While TOTALAMOUNT is greater than or equal to 500, increment NOOFFIVE by 1 and subtract 500 from TOTALAMOUNT.

Step 7: While TOTALAMOUNT is greater than or equal to 200, increment NOOFTWO by 1 and subtract 200 from TOTALAMOUNT.

Step 8: While TOTALAMOUNT is greater than or equal to 100, increment NOOFHUN by 1 and subtract 100 from TOTALAMOUNT.

Step 9: Display the number of ■500, ■200, and ■100 notes.

Step 10: Stop the program.

4. SAMPLE OUTPUT

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ENTER THE TOTAL AMOUNT
1200
500: 2
200: 1
100: 1
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Note: The sample output assumes the counters are initialized to zero before the calculations. In COBOL, initialize NOOFFIVE, NOOFTWO, and NOOFHUN to zero (for example, with VALUE 0) to ensure correct results.